

# ECON 695: Practice Problem Set

Two-Tier Compliance (LATE)

## Practice Questions

November 11, 2025

### Problem: The Smoking Cessation Program

A study evaluates a new program to help pregnant smokers quit. 3,000 pregnant smokers are randomly assigned to a treatment or control group.

- Individuals in the **treatment group** ( $Z_i = 1$ ) receive a voucher that pays for a 3-month specialized smoking cessation class.
- Individuals in the **control group** ( $Z_i = 0$ ) receive no voucher but are given a standard informational pamphlet.

The researchers collect data on:

- $Z_i$ : Treatment assignment (1 = Voucher, 0 = Control).
- $D_{1i}$ : Whether the individual **attended the class** (1 = Yes, 0 = No).
- $D_{2i}$ : Whether the individual **quit smoking** by the third trimester (1 = Yes, 0 = No).
- $Y_i$ : The birthweight of their baby (in grams).

### Key Assumptions

- **Monotonicity:** The voucher ( $Z_i$ ) makes people \*more\* likely to attend the class ( $D_{1i}$ ) and to quit smoking ( $D_{2i}$ ). There are no defiers for either action.
- **Exclusion Restriction:** The voucher ( $Z_i$ ) only affects birthweight ( $Y_i$ ) \*through\* its effect on class attendance ( $D_{1i}$ ) or quitting ( $D_{2i}$ ).
- **Program Structure:** An individual can only quit smoking \*as a result of the program\* if they first attend the class. (Note: Some people might quit on their own without the class, but the program's \*effect\* on quitting is channeled through the class).

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### Questions

#### 1. Defining Compliance Types

An individual's 'type' is defined by their potential behaviors under both treatment and control. In this case, the behavior is a *pair* of actions: (Class Attendance, Quitting Status).

Let  $(D_{1i}(z), D_{2i}(z))$  be the pair of actions an individual takes given instrument status  $z$ . A person's type is defined by the full vector:  $(D_{1i}(0), D_{2i}(0), D_{1i}(1), D_{2i}(1))$ .

Assume for simplicity that (a) no one quits \*due to the program\* without attending the class and (b) there are no defiers.

- (a) Create a table summarizing the possible compliance types. The columns should be: "Type," "Behavior if Control:  $(D_1(0), D_2(0))$ ," "Behavior if Treatment:  $(D_1(1), D_2(1))$ ," and "Description."  
*(Hint: Start with the simplest types like 'Never-Taker' and 'Always-Taker'. Then, consider the "two-tiers" of compliers your professor mentioned. What does a person who complies with the class but not with quitting look like? What about a person who complies with both?)*

## 2. Population Proportions

You are given the following observed probabilities (population expectations) from the data:

| Table 1: Observed Probabilities by Treatment Status |                             |                           |
|-----------------------------------------------------|-----------------------------|---------------------------|
|                                                     | $E[D_1 Z]$ (Attended Class) | $E[D_2 Z]$ (Quit Smoking) |
| <b>Treatment</b> ( $Z = 1$ )                        | 0.60                        | 0.30                      |
| <b>Control</b> ( $Z = 0$ )                          | 0.10                        | 0.10                      |

- (a) What is the proportion of '**Class-Compliers**' in the population? (This is the group induced to attend class by the voucher, regardless of whether they quit).
- (b) What is the proportion of '**Quit-Compliers**' in the population? (This is the group induced to quit smoking by the voucher).

### 3. Interpreting LATE

You are also given the following average outcomes:

- $E[Y_i|Z_i = 1] = 3100\text{g}$  (Average birthweight for voucher group)
- $E[Y_i|Z_i = 0] = 3000\text{g}$  (Average birthweight for control group)

Suppose you run two separate IV (Wald) estimates.

- (a) **Estimate 1:** You use  $Z_i$  as an instrument for  $D_{1i}$  (class attendance). What LATE does this estimate? Calculate its value and interpret it in words.

$$\text{LATE}_{D_1} = \frac{E[Y|Z = 1] - E[Y|Z = 0]}{E[D_1|Z = 1] - E[D_1|Z = 0]}$$

- (b) **Estimate 2:** You use  $Z_i$  as an instrument for  $D_{2i}$  (quitting smoking). What LATE does this estimate? Calculate its value and interpret it in words.

$$\text{LATE}_{D_2} = \frac{E[Y|Z = 1] - E[Y|Z = 0]}{E[D_2|Z = 1] - E[D_2|Z = 0]}$$

- (c) **Comparison:** Why are these two LATE estimates different? Which specific group(s) of compliers (from your table in Q1) does Estimate 2 (the 'Quit-Complier' LATE) apply to?